

Portfolio Flood Risk Assessment: Key Formulae

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1 Property Valuation Formulae

1.1 Annualized Growth Rate

$$\text{AGR} = \left(\frac{V_{\text{current}}}{V_{\text{purchase}}} \right)^{\frac{1}{t}} - 1$$

where t is years since purchase

1.2 Local Price Index

$$\text{LPI}_i = \text{median}\{V_j : d(i, j) \leq r\}$$

where $d(i, j)$ is distance between properties i and j , and r is radius

2 Flood Risk Formulae

2.1 Flood Depth Calculation

$$D_i = \max\left(0, D_{\text{max}} \left(1 - \frac{d_i}{R}\right)\right)$$

where:

- D_i is flood depth at property i
- D_{max} is maximum flood depth
- d_i is distance to flood center
- R is flood radius

2.2 Spatial Correlation

$$\rho_{ij} = \rho_0 \exp\left(-\frac{d_{ij}}{d_c}\right)$$

where:

- ρ_0 is base correlation
- d_{ij} is distance between properties
- d_c is correlation distance

2.3 Impact Calculation

$$I_i = V_i \cdot \alpha(1 + \tanh(D_i))$$

where:

- I_i is impact on property i
- V_i is property value
- α is baseline discount
- D_i is flood depth

3 Portfolio Metrics

3.1 Geographic Concentration (HHI)

$$\text{HHI} = \sum_{i=1}^n \left(\frac{V_i}{\sum_{j=1}^n V_j} \right)^2$$

3.2 Expected Shortfall

$$\text{ES}_\alpha = \mathbb{E}[X|X > \text{VaR}_\alpha]$$

where X is portfolio impact